Course Notes: AI Agents

An AI agent combines a language model, tools, and an execution loop.

Tools provide access to information outside the model knowledge.

Example: read a local PDF and save a study outline.

A focused agent has one clear purpose and a small set of tools.

Function Calling and Evidence

Function Calling lets the model request a named tool with JSON arguments.

The application validates arguments and executes the requested function.

The tool result is sent back to the model for the next decision.

A study assistant should cite PDF page numbers and report missing evidence.

It must report saving success only after the save tool succeeds.